

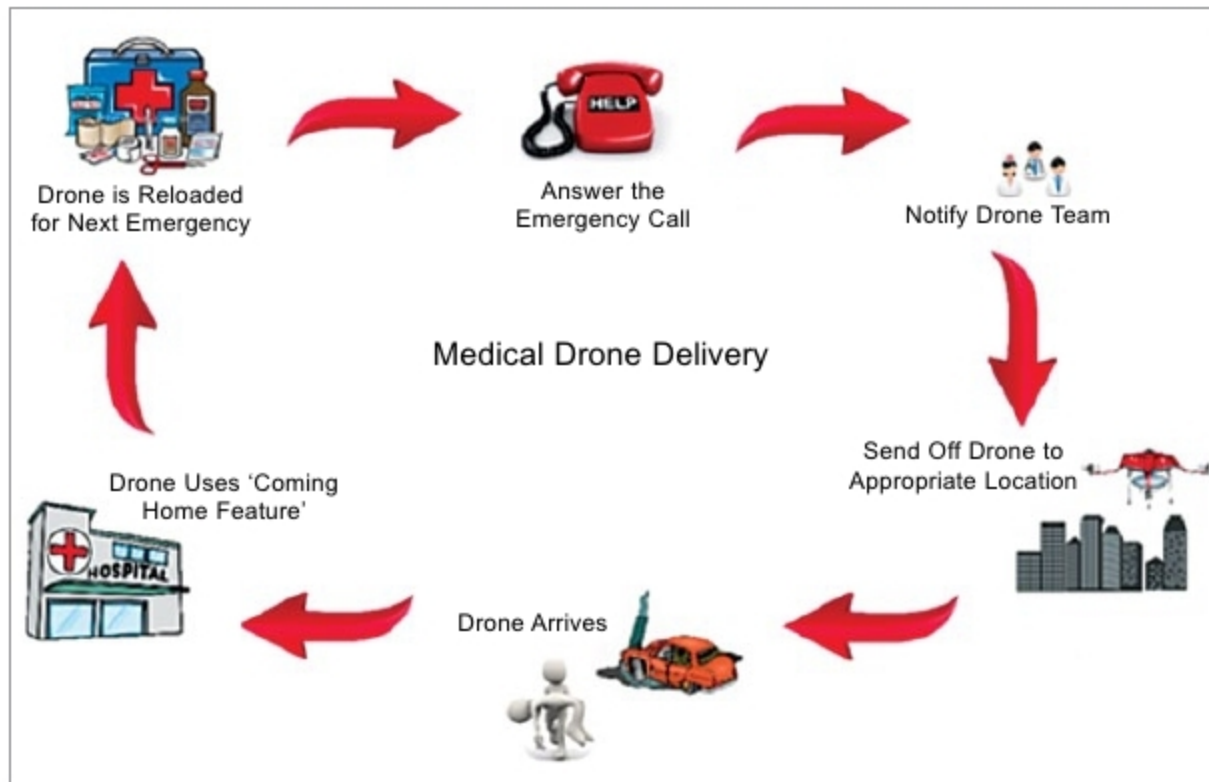


Fig. 2: Workflow of the drone ambulance

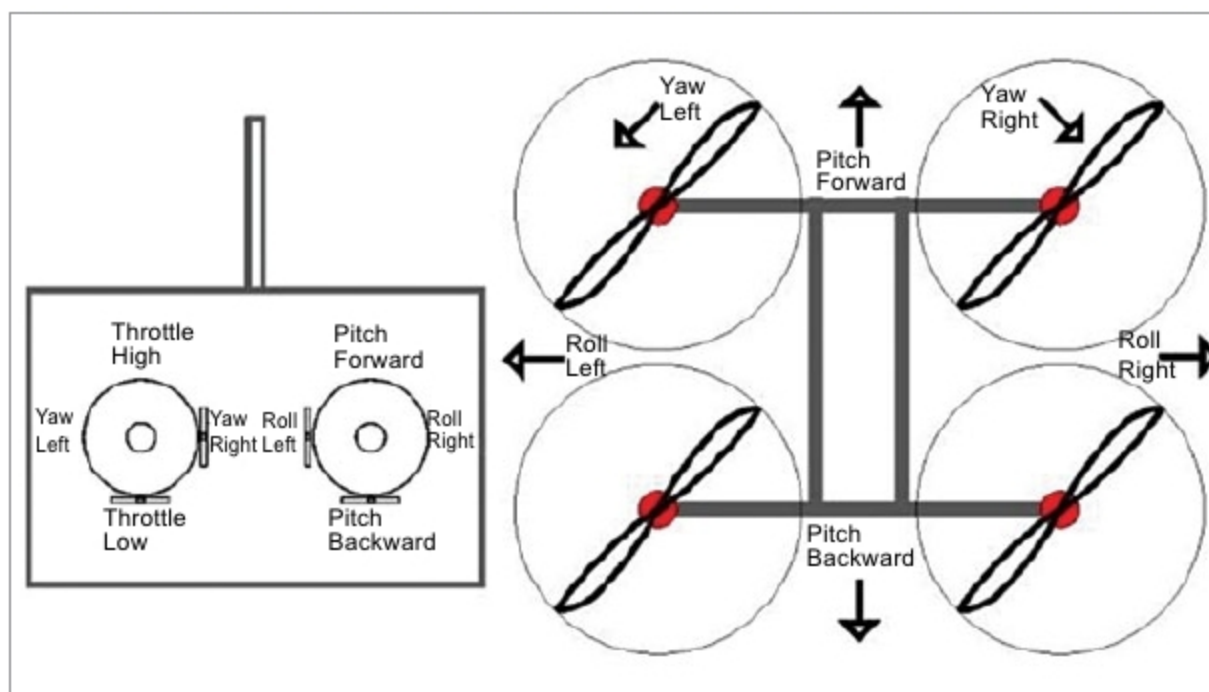


Fig. 3: Throttle stability in the quadcopter (Credit: www.uavcoach.com)

BLDC motors on each of the rails and then integrate four electronic speed controllers (ESCs) on each rail, and connect the common positive and negative terminals of the four ESCs to the respective positive and negative terminals of the battery.

After this, propellers were incorporated to the motors with the aim to move two diagonally-opposite propellers in clockwise direction and the remaining two in anti-clockwise direction. This marked the end of the quadcopter fabrication. A six-channel IR remote was used to fly the drone.

Incorporating the medbox. The medbox comprises other internal circuitry such as Arduino Uno, ECG sensor, heartbeat sensor, thermistor and other medical requirements like

some tablets. It also contains a user manual through which a person can observe and help the patient in need. Architecture of the quadcopter is shown in Fig. 1. Power is distributed via power distribution board among the flight controller, sensors as well as the four ESCs. The four ESCs drive each of the four motors.

Workflow

Consider a situation in which a person suffers from a sudden cardiac arrest. When the hospital is informed of this incident, an ambulance is immediately sent to the emergency spot. Simultaneously, the drone team is also notified. The drone is then sent off to the target location. It is well-equipped with the medbox, which consists of

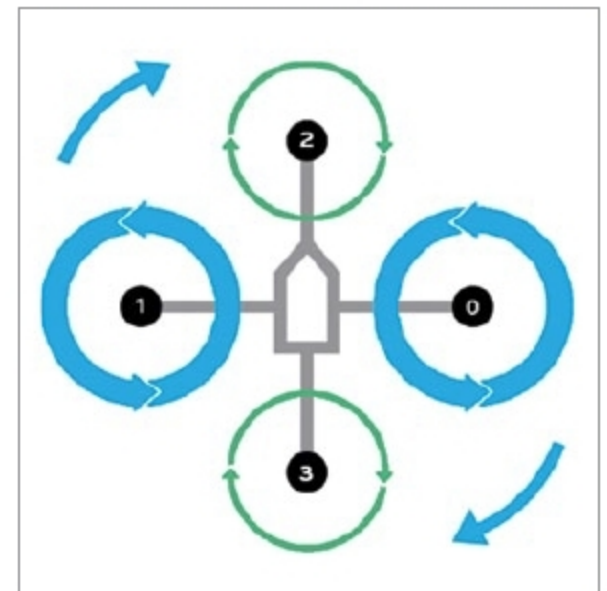


Fig. 4: Yaw stability in the quadcopter (Credit: www.uavcoach.com)

various sensors and the first aid kit.

Measured real-time health parameters of the patient are transmitted via ZigBee to the ambulance. These parameters can be observed and analysed in LabView software. Once the ambulance reaches the emergency spot, the drone is sent back to the hospital and is reloaded for the next operation.

Quadcopter specifications

A quadcopter, multirotor, drone or quadrotor is a simple flying mechanical vehicle that has four arms, and in each arm, there is a motor attached to a propeller. Each rotor produces both a thrust and torque about its centre of rotation, as well as a drag force opposite to the vehicle's direction of flight. Similar to other flying objects, a quadcopter has a group of forces and torques acting on it while it flies. There are four main forces acting on the drone: drag, lift, weight and thrust. For the drone to fly, these different forces need to be balanced.

Given below are some common motion controls for a quadcopter:

Throttle. Throttle adjusts the altitude, or height, of the quadcopter. A quadrotor hovers or adjusts its altitude by applying equal thrust to all four rotors. On most quadcopters, throttle is done by pushing the left stick up (to go up) or down (to go down).

Yaw. Yaw is the rotation of the head of the quadcopter, either right or left. It helps with changing directions while flying. A quadrotor adjusts its yaw by applying more thrust to rotors rotating in one direction. On most quadcopters, yaw is controlled by